1 Single Winner Case

1. Consider the single winner case when the jackpot amount is $\$1.15 \times 10^9$. The amount the winner would receive varies based on whether the jackpot is paid as a lump sum or annuity, and after taxes:

   - Lump sum payout is typically about 60% of the advertised jackpot
   - We also need to subtract Federal taxes, which are in the 24-37% range

   Then for a single winner of a $1.15B jackpot:

   - The lump sum gives you something like 60% of the jackpot
     - So assume that the lump sum is $(1.15 \times 10^9) \times 0.60 = \$690,000,000$
   - Federal tax is in the 24-37% range, so for simplicity assume a tax rate of 30%
     - Then after taxes we have $\$690,000,000 \times (1.00 - 0.30) = \$483,000,000$

   So we'll use $483,000,000 as the effective jackpot.

2. Now we can calculate the expected value of the jackpot, $\text{EV}_{\text{jackpot}}$ as follows:

$$\text{EV}_{\text{jackpot}} = \frac{1}{302,575,350} \times \$483,000,000 \approx \$1.60$$

3. The expected value of a $2 ticket (for the jackpot only) is approximately $1.60 when the jackpot is $\$1.15 \times 10^9$ and there is a single winner.

Summary: For a $1.15 billion jackpot, the jackpot-only expected value of a $2 Mega Millions ticket when there is a single winner is approximately $1.60 before considering the cost of a ticket. After subtracting the $2 ticket cost, the net expected value is apparently (0.40), or -40 cents.

2 Multiple Winners Case

These are the key values for this case:

1. Advertised Jackpot: $\$1.15 \times 10^9$
2. Lump Sum Option: Typically 60% of the advertised amount, so take
   Lump sum $= (\$1.15 \times 10^9) \times 0.60 = \$690 \times 10^6$
3. Taxes: Both Federal and potentially state taxes apply
   - Federal tax rate: 37%
   - State tax rates vary but average around 5%
   - Combined tax rate: 42%
   - Post-tax lump sum $= \$690 \times 10^6 \times (1.00 - 0.42) = \$400.2 \times 10^6$.

2.1 Adjusted Lump Sum per Winner

If the jackpot is split among n winners, each winner receives:

$$\text{Post-Tax Lump Sum Share} = \frac{\text{Post-Tax Lump Sum}}{n} = \frac{\$400.2 \times 10^6}{n}$$

For example:

- $n = 1$: $\$400.2 \times 10^6$
- $n = 2$: $\$200.1 \times 10^6$
- $n = 3$: $\$133.4 \times 10^6$

2.2 Expected Number of Winners

The expected number of winners depends on T, the number of tickets sold, and the odds of winning as follows

$$\text{Expected Number of Winners} = T \times \frac{1}{302,575,350}$$

Suppose 500 million tickets are sold, so that $T = 500 \times 10^6$. Then we have

$$\text{Expected Number of Winners} = (500 \times 10^6) \times \frac{1}{302,575,350} = 1.65$$

2.3 Post-Tax Lump Sum per Winner

If there are 1.65 winners on average, each winner receives:

$$\text{Post-Tax Lump Sum Share} = \frac{\$400.2 \times 10^6}{1.65} = \$242.5 \times 10^6$$

2.4 Expected Jackpot Value per Ticket

The expected jackpot payout per ticket is:

$$\text{EV}_{\text{jackpot}} = \text{Post-Tax Lump Sum Share} \times \text{Odds of Winning}$$

So $\text{EV}_{\text{jackpot}}$ is

$$\text{EV}_{\text{jackpot}} = (\$242.5 \times 10^6) \times \frac{1}{302,575,350} \approx \$0.80$$

2.5 Net $\text{EV}_{\text{jackpot}}$

$$\text{EV}_{\text{jackpot}} = 0.80 - 2 = -\$1.20$$

Summary: For a $\$1.15 \times 10^9$ jackpot with 500×10^6 tickets sold, the expected value of the jackpot portion only is approximately \$0.80. Subtracting the ticket cost, the net EV is approximately (1.20), meaning the ticket is not profitable in this scenario.

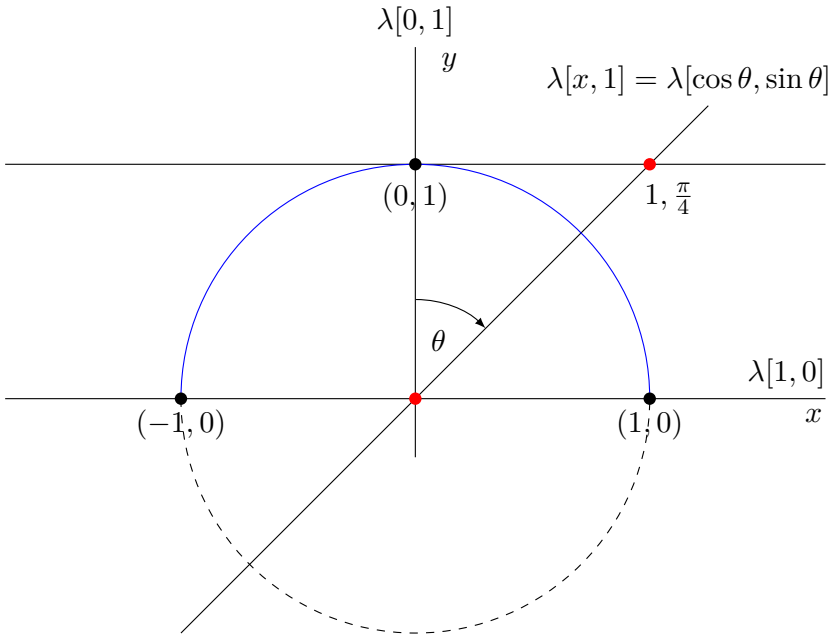

Figure 1: Real Projective Line Setup

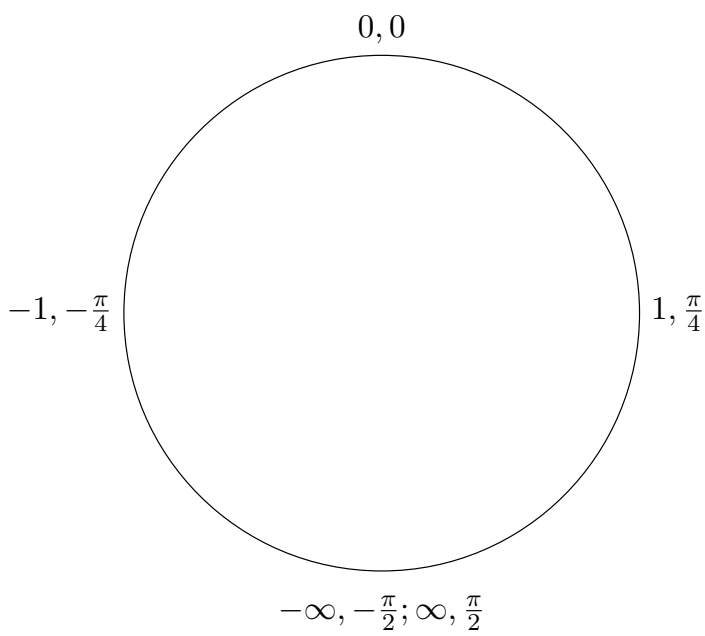

Figure 2: The Real Projective Line is a Circle: (x, θ)

What	Linear Algebra	Fourier Series
Vector Space	$\mathbb{R}^n$	Piecewise smooth 2π-periodic functions on $\mathbb{R}$
Inner Product	$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^{n} u_i v_i$	$\langle f(t), g(t) \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) g(t)\, dt$
Orthonormal Basis $\left(\mathbb{R}^3\right)$	$\{(1,0,0),(0,1,0),(0,0,1)\}$	$\{1, \cos mt, \sin nt\},\ n, m \in \mathbb{N} \backslash \{0\}$
Representation of a Vector in the Basis	$\mathbf{x} = \sum_{i=1}^{n} a_i \mathbf{e}_i$	$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt$
Coefficients are Projections	$a_i = \langle \mathbf{x}, \mathbf{e}_i \rangle$	$a_0 = \langle f(t), 1 \rangle$ $a_m = \langle f(t), \cos mt \rangle$ $b_m = \langle f(t), \sin mt \rangle$

Table 1: Vector Spaces: Linear Algebra vs. Fourier Series

Note: Some people think that the orthonormal basis for the Fourier series should be $\{\mathbf{1}, \sqrt{2}\cos mt, \sqrt{2}\sin nt\}$, since

$$\langle \mathbf{1}, \mathbf{1} \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} \mathbf{1}^2\, dt = \frac{1}{\pi} \int_{-\pi}^{\pi} dt = \frac{1}{\pi}\, t \Big|_{-\pi}^{\pi} = \frac{1}{\pi}\left(\pi - (-\pi)\right) = \frac{1}{\pi} 2\pi = 2$$

OTOH, $\langle \cos nt, \cos nt \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos^2 nt\, dt = \frac{1}{\pi}\pi = 1$. In the same way, $\langle \sin nt, \sin nt \rangle = 1$.

One of the points here (or perhaps, the point here) is that $\langle \hat{\mathbf{u}}, \hat{\mathbf{u}} \rangle = 1$ for all unit vectors $\hat{\mathbf{u}}$, so what about $\langle \mathbf{1}, \mathbf{1} \rangle$ if $\mathbf{1}$ is part of an orthonormal basis (again, since $\langle \mathbf{1}, \mathbf{1} \rangle = 2$)?

Note also that here $\langle \mathbf{1}, \sin nt \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} \mathbf{1} \sin nt\, dt = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin nt\, dt = 0$. Similarly, $\langle \mathbf{1}, \cos nt \rangle = \langle \cos nt, \sin mt \rangle = 0$.

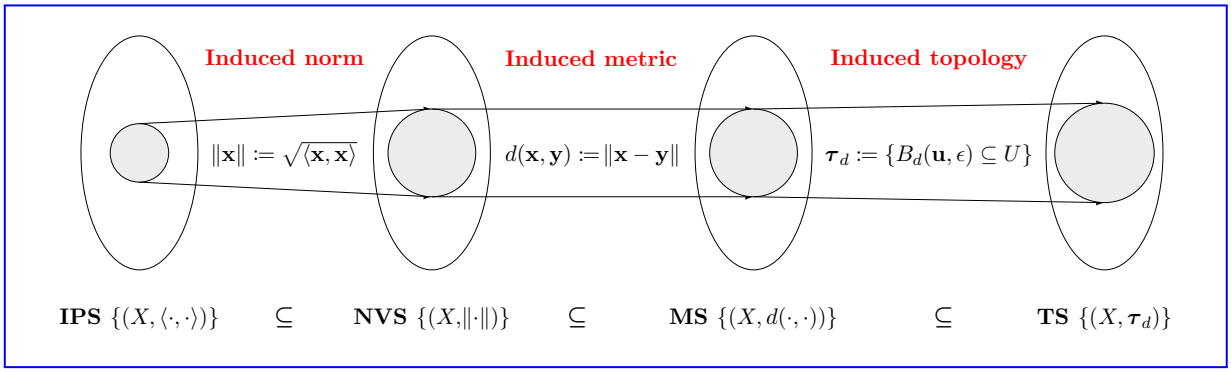

Figure 3: Induced Norm, Metric, and Topology

IPS	$--$	Inner Product Spaces	$(X, \langle \mathbf{x}, \mathbf{y} \rangle)$
NVS	$--$	Normed Vector Spaces	$(X, \|\mathbf{x}\|) = (X, \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle})$
MS	$--$	Metric Spaces	$(X, d(\mathbf{x}, \mathbf{y})) = (X, \|\mathbf{x} - \mathbf{y}\|) = (X, \sqrt{\langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y} \rangle})$
TS	$--$	Topological Spaces	$(X, \boldsymbol{\tau}_d)$, where $\boldsymbol{\tau}_d = \{\text{open subsets of } (X, d)\}$

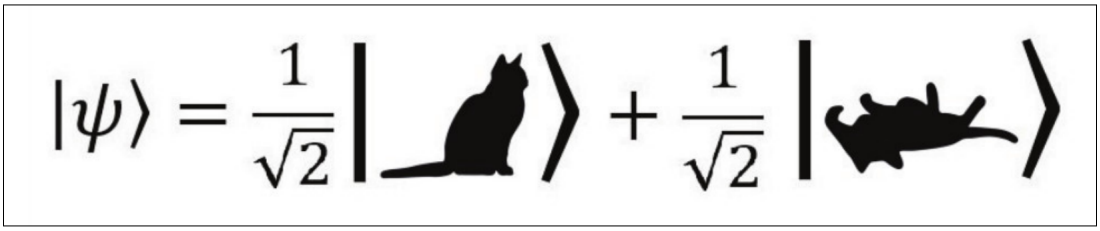

Figure 4: Quantum superposition and Schrödinger's cat

$n=\langle\ ,\ \rangle$
$G(\ )$
$\ \ \ \

0.6249999121401877

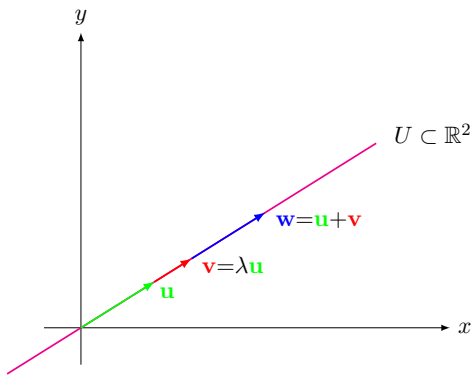

Figure 7: The line U is a linear subspace of $\mathbb{R}^2$

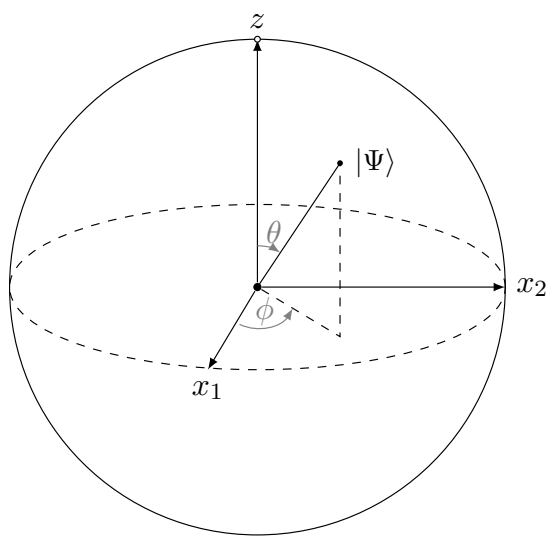

Figure 8: The Bloch Sphere

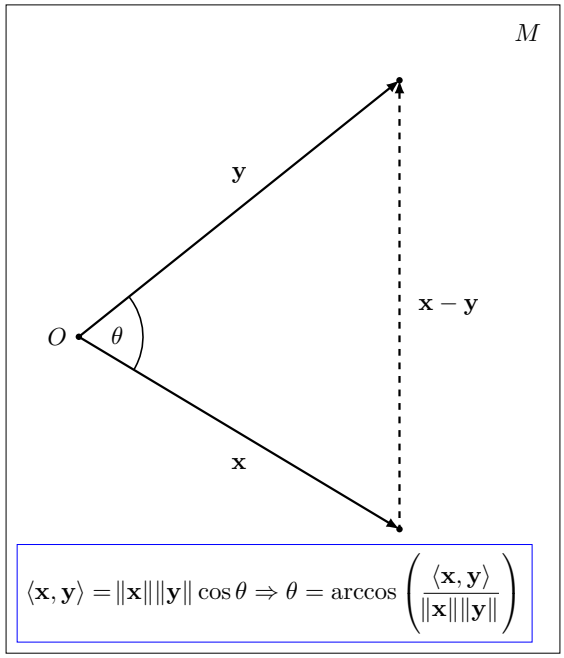

Figure 9: Geometric interpretation of $\langle \mathbf{x}, \mathbf{y} \rangle$

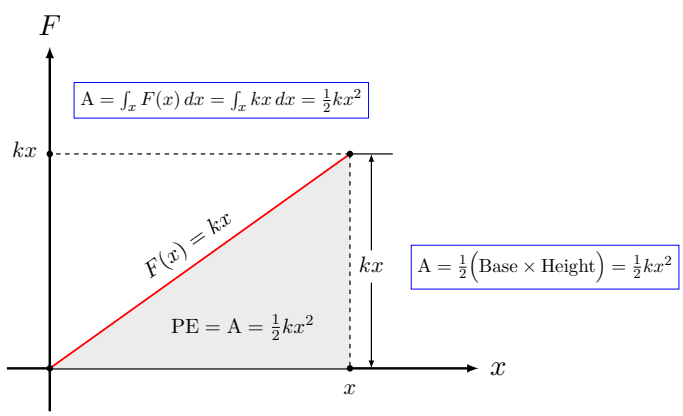

Figure 10: Potential Energy of a Simple Harmonic Oscillator

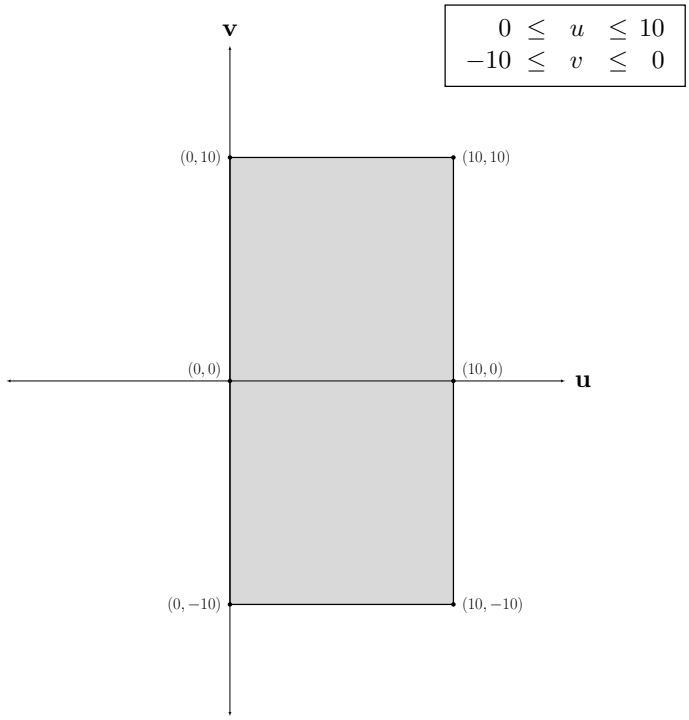

Figure 11: The transformed parallelogram is a rectangle in the uv-plane

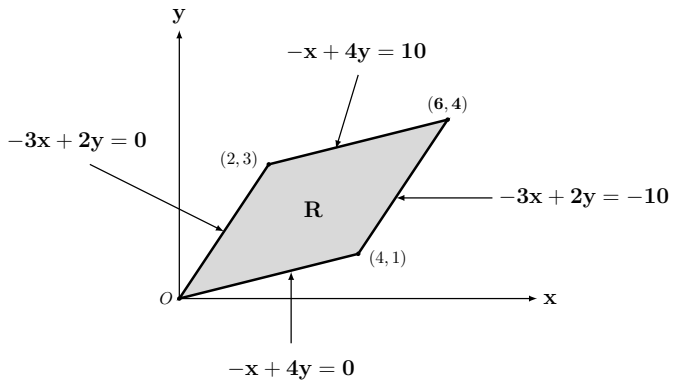

Figure 12: Equations of the Sides of the Parallelogram

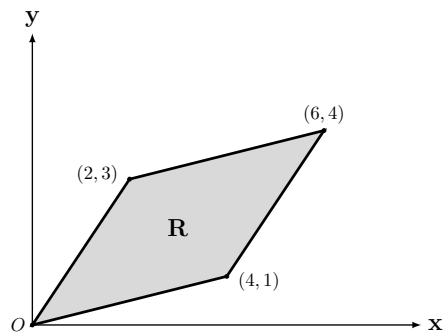

Figure 13: Parallelogram in the xy-plane

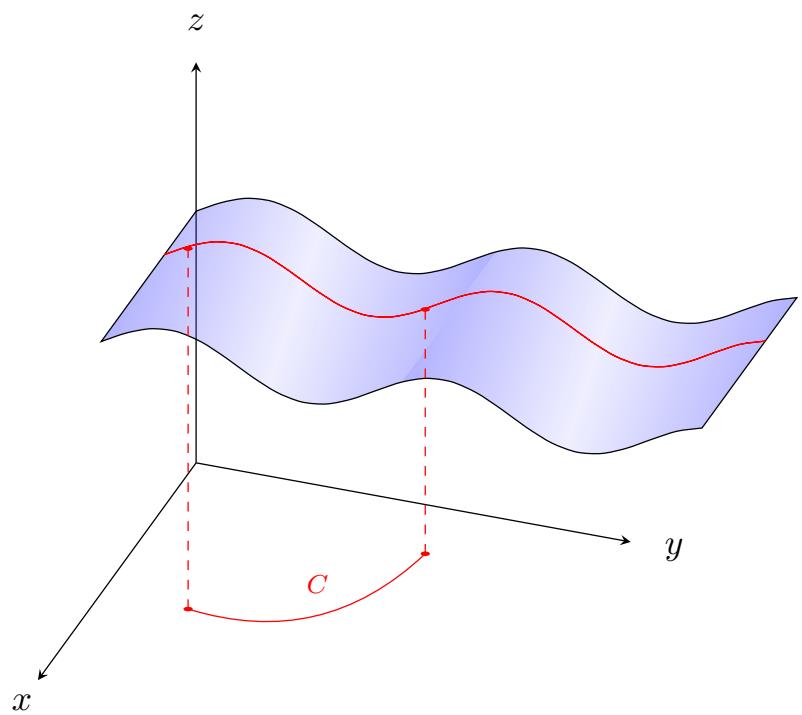

Figure 14: 3D

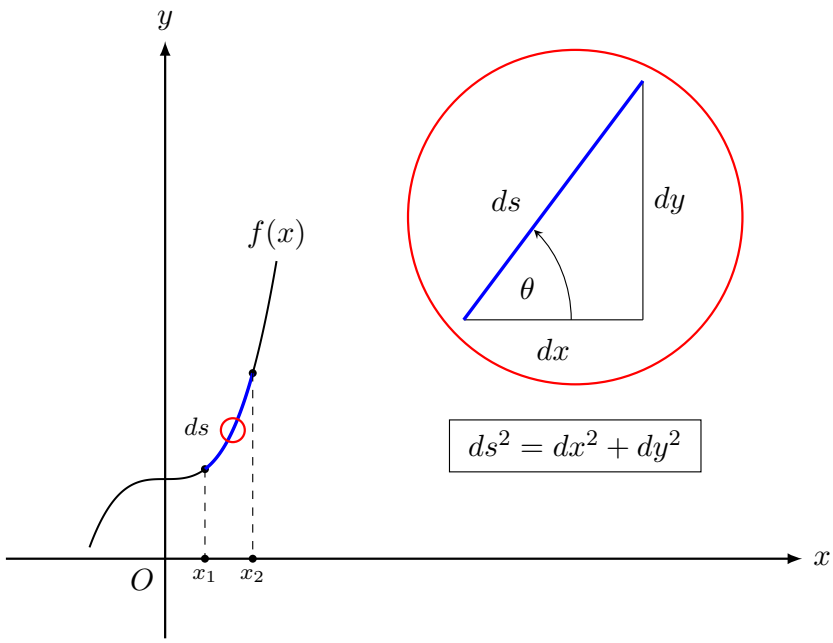

Figure 15: $f(x)$, ds and the Pythagorean Theorem

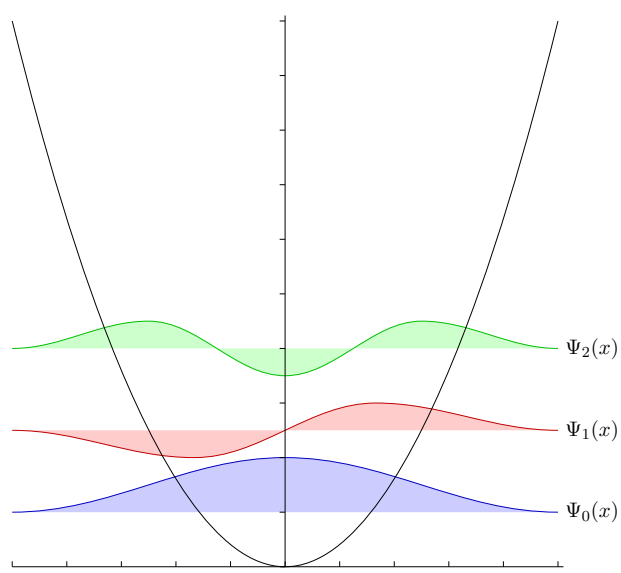

Figure 16: Wave Functions

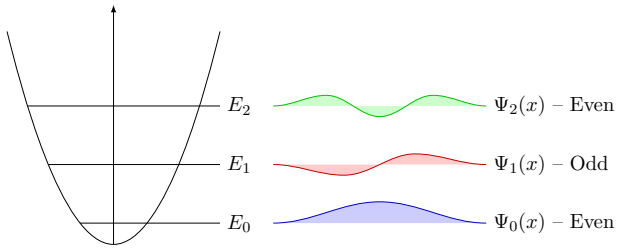

Figure 17: Wave Functions

9^2	$=$	81	$\longrightarrow$	$8+1$	$=$	9
45^2	$=$	2025	$\longrightarrow$	$20+25$	$=$	45
703^2	$=$	494209	$\longrightarrow$	$494+209$	$=$	703
7777^2	$=$	60481729	$\longrightarrow$	$6048+1729$	$=$	7777
857143^2	$=$	734694122449	$\longrightarrow$	$734694+122449$	$=$	857143

Figure 18: A Few Example Kaprekar Numbers

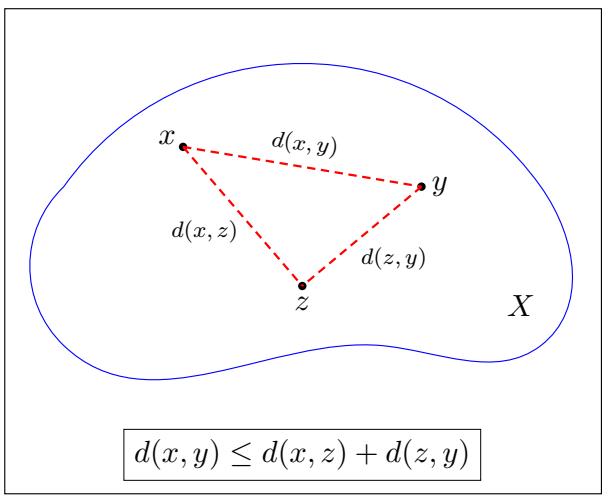

Figure 19: The Triangle Inequality

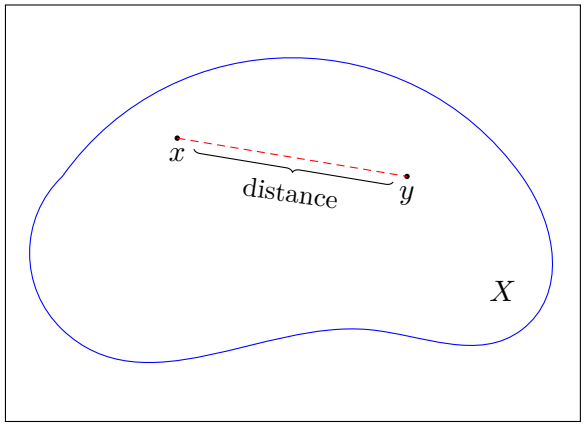

Figure 20: X, x and y

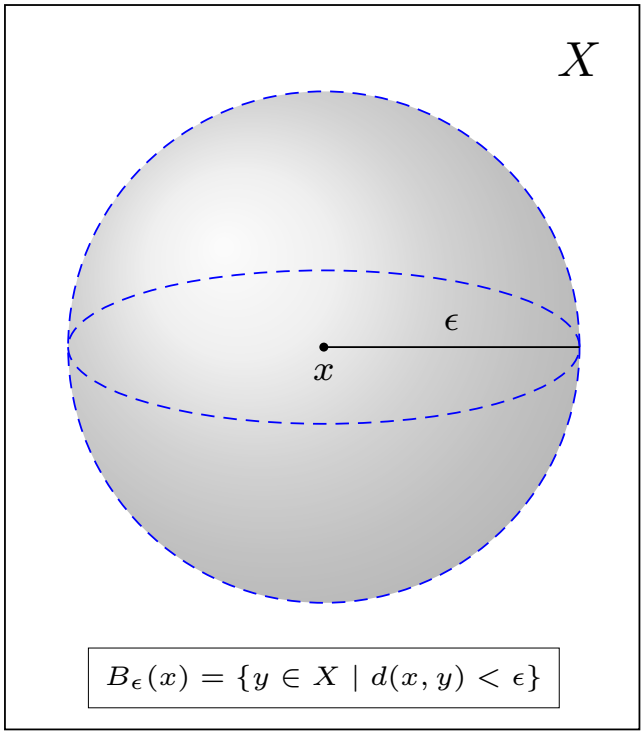

Figure 21: $B_\epsilon(x)$ is an open epsilon ball centered at x with radius ϵ

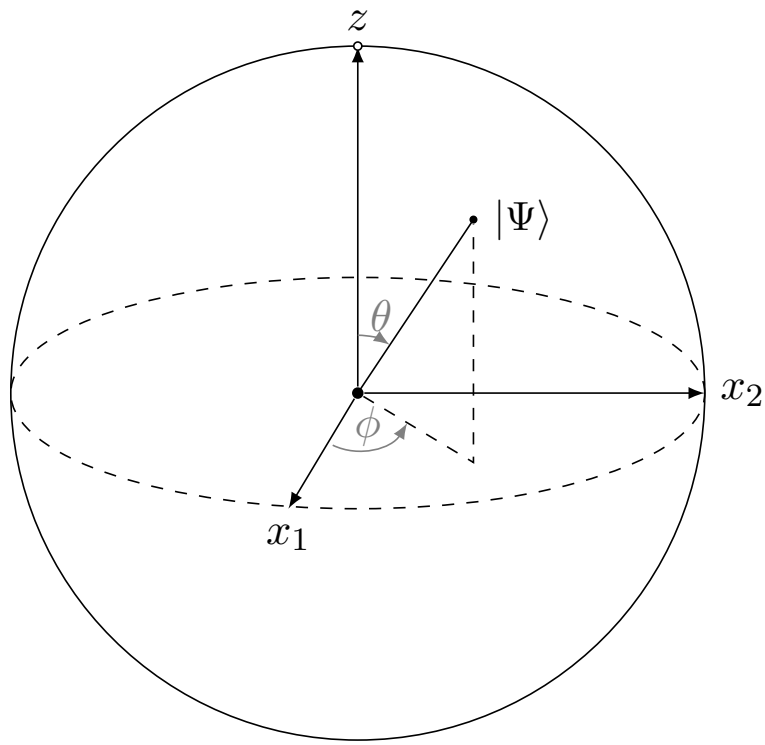

Figure 22: The Bloch Sphere

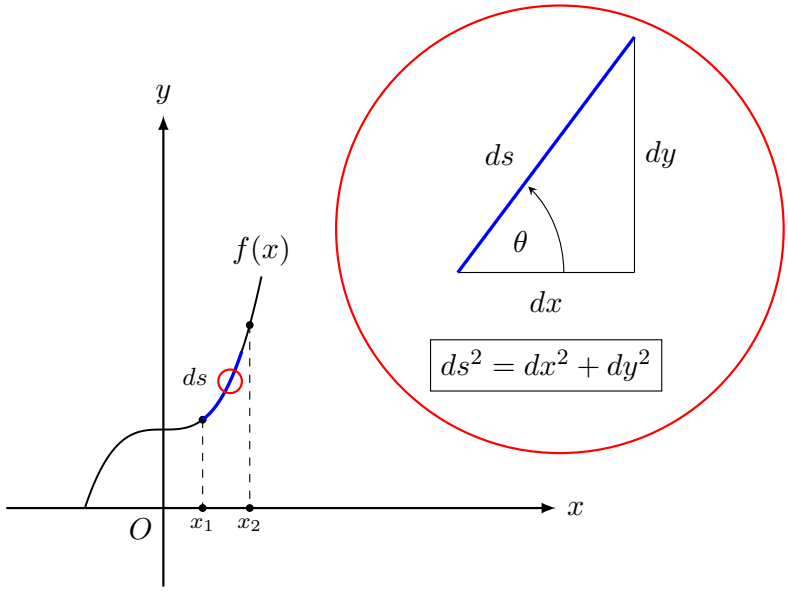

Figure 23: $f(x)$, ds and the Pythagorean Theorem

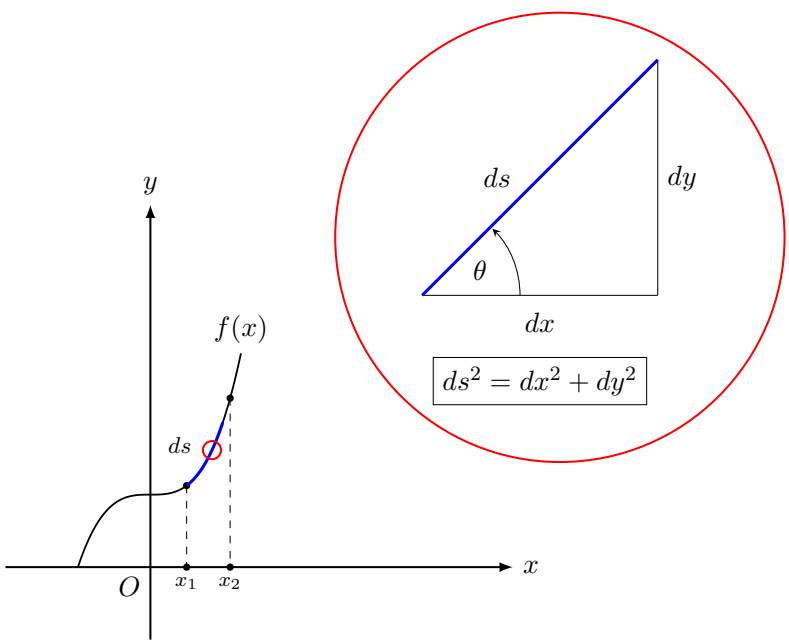

Figure 24: $f(x)$, ds and the Pythagorean Theorem